

05 — AI Assistant Specification

AI provider: Gemini or OpenAI API later. MVP must work without AI.

AI Role

- AI is an optional explanation and education layer, not the calculation engine.
- Core cycle, fertile-window and pregnancy-date calculations are deterministic application logic.
- AI may explain the user's tracked data, summarize patterns, explain general educational content and help prepare questions for a clinician.

Provider Abstraction

- Implement an `AIService` interface so Gemini and OpenAI can be swapped without changing the UI.
- Suggested interface: `sendMessage(context, userQuestion) → structured response`.
- Keep API keys out of the Flutter client in any production deployment. For a local prototype, document the limitation and use environment variables; later move requests behind a secure backend.

System Safety Rules

- Never diagnose.
- Never claim certainty about pregnancy, ovulation or medical conditions.
- Never prescribe or change medication.
- Never invent sources.
- When medical risk is possible, recommend professional medical assessment rather than reassurance.
- Use the user's stored data only when necessary for the question.
- Do not expose private partner data unless sharing permission explicitly allows it.

AI Context

- Send only the minimum required structured context: current mode, relevant dates, calculated status, user-entered observations and selected educational article context.
- Do not send the entire database or unrelated personal information.
- Separate confirmed user observations from app estimates in the prompt.

Example Tasks

- Explain: 'What does cycle day 13 mean?'
- Summarize: 'What patterns do my last 6 cycles show?'
- Prepare: 'What should I ask my doctor about irregular cycles?'
- Pregnancy education: 'Explain what the app means by week 10.'
- Never answer: 'Do I have PCOS?' with a diagnosis.

Offline/Fallback

- If AI is unavailable, the app still provides all core tracking, calculations and static educational content.

- Show a normal network/error state rather than pretending the AI answered.

Provider sources

- Gemini API documentation should be checked at implementation time: <https://ai.google.dev/>
- OpenAI API documentation should be checked at implementation time: <https://platform.openai.com/docs/>